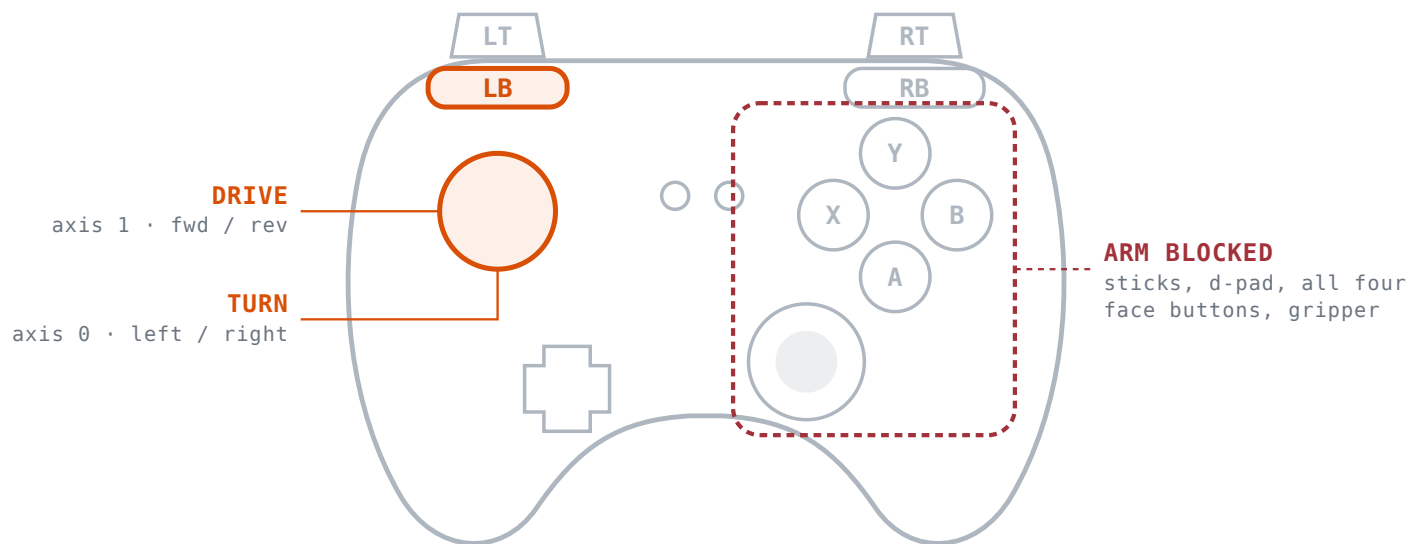




Only the left stick is live. Holding LB hard-stops the arm before anything else is read.

LIVE CONTROLS

- L stick ▲▼ Drive forward / reverse**
axis 1 · 0.55 m/s mock, 0.45 m/s 0Drive
- L stick ◀▶ Turn**
axis 0 · 1.60 rad/s mock, 1.40 rad/s
0Drive

INERT WHILE LB IS HELD

- Everything** Right stick, d-pad, A / B / X / Y and the whole arm do nothing.
- Drill too** LB + LT reaches nothing. LB wins over the drill's gate as well.

The arm is actively blocked, not just idle. The teleop node checks LB first, commands a hard zero stop, and returns before reading any other control — it logs `LB rover mode active: arm blocked`. Stick deadzone here is 0.08, wider than the arm's 0.04.

NO MODIFIER

Nothing moves. Teleop publishes **silence**, not zeros.

BOTH RB + RT

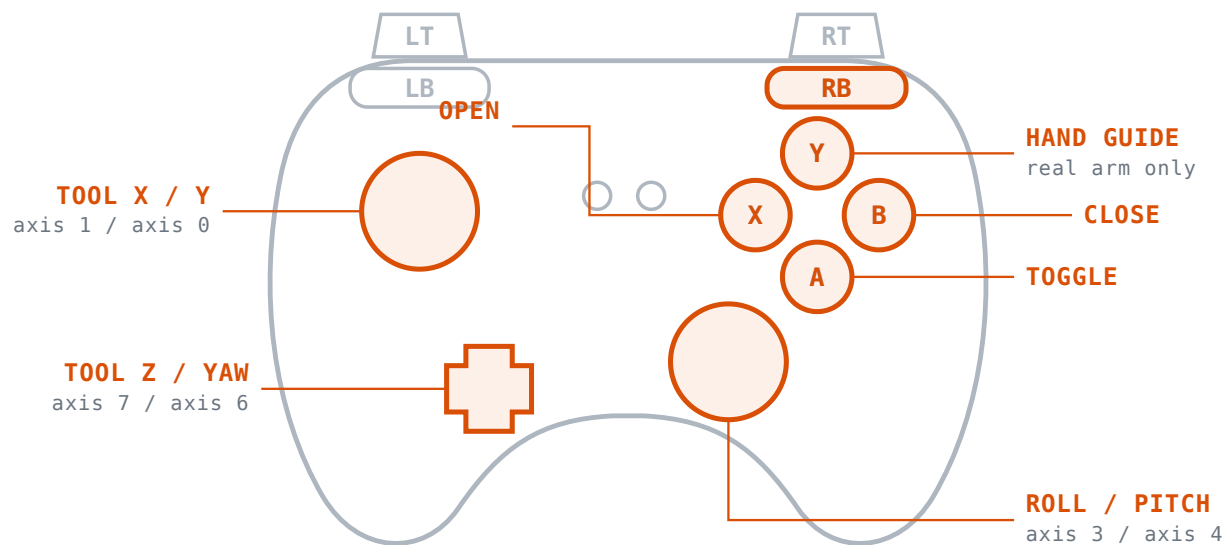
RB wins. A brushed trigger can't change the sticks.

TRIGGER THRESHOLD

LT / RT engage past **0.50**, release below **0.35**.

SIGNAL LOSS

/joy silent **0.35 s** → arm stops.



Translation on the left stick, rotation on the right, Z and yaw on the d-pad. The whole face cluster is gripper work, except Y.

TRANSLATE — 0.10 M/S AT FULL DEFLECTION

L stick ▲▼ **Tool X** axis 1
L stick ◀▶ **Tool Y** axis 0
D-pad ▲▼ **Tool Z** axis 7

ROTATE — 0.18 RAD/S, ABOUT 10°/S

R stick ◀▶ **Roll** axis 3
R stick ▲▼ **Pitch** axis 4
D-pad ◀▶ **Yaw** axis 6

GRIPPER & HAND GUIDING

X · hold **Open** 2.0 rad/s, full stroke ~0.82 s
B · hold **Close**
A · press **Toggle** open ↔ closed
Y · hold **Hand guiding** — ZeroTorque, real arm only

Release overshoot is real. The arm keeps travelling roughly 0.25 s after you let go — transport delay plus the ReBeL's own firmware deceleration, neither reachable from ROS. Release slower to stop shorter.

NO MODIFIER

Nothing moves. Teleop publishes **silence**, not zeros.

BOTH RB + RT

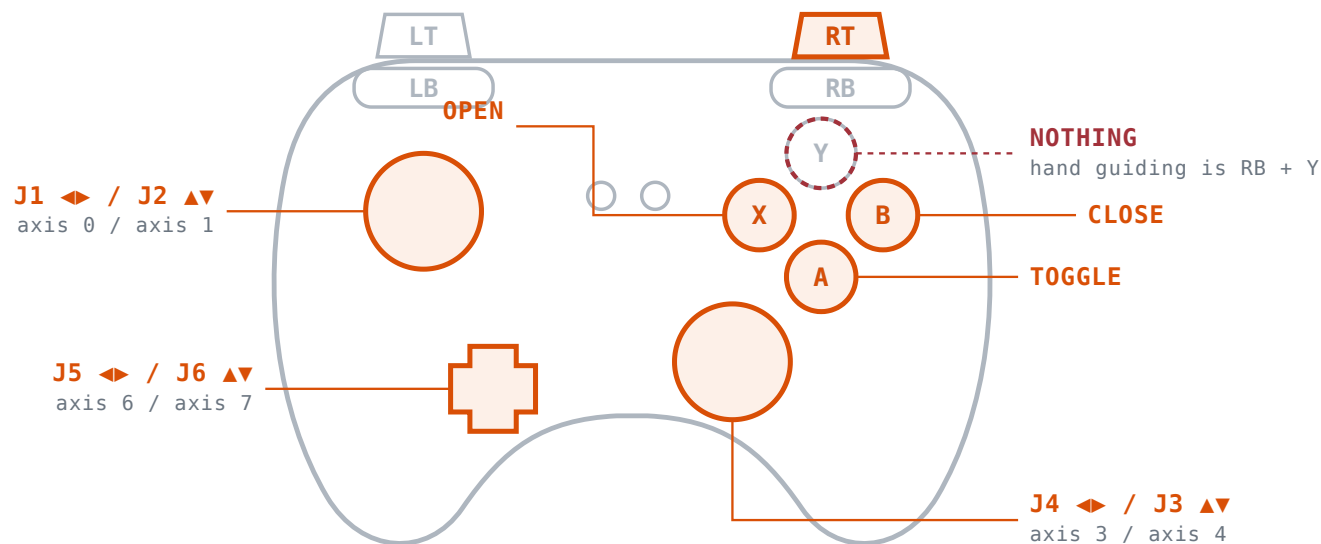
RB wins. A brushed trigger can't change the sticks.

TRIGGER THRESHOLD

LT / RT engage past **0.50**, release below **0.35**.

SIGNAL LOSS

/joy silent **0.35 s** → arm stops.



Same sticks, different meaning: six joints instead of a tool frame. Y goes dead here — hand guiding needs RB.

STICKS — 0.38 RAD/S PER JOINT

L stick ◀ **joint1** axis 0 · base yaw
L stick ▲ **joint2** axis 1 · shoulder
R stick ▲ **joint3** axis 4 · elbow
R stick ◀ **joint4** axis 3 · forearm roll

D-PAD — THE WRIST

D-pad ◀ **joint5** axis 6 · wrist pitch
D-pad ▲ **joint6** axis 7 · wrist roll

GRIPPER — IDENTICAL TO RB

X · hold **Open**
B · hold **Close**
A · press **Toggle**
Y Nothing. Hand guiding is RB + Y, never RT + Y.

0.28 rad/s is a ceiling, not the speed. If any joint exceeds it the whole velocity vector scales down uniformly, so direction is preserved — but near a singularity the tool moves visibly slower than the number suggests.

NO MODIFIER

Nothing moves. Teleop publishes **silence**, not zeros.

BOTH RB + RT

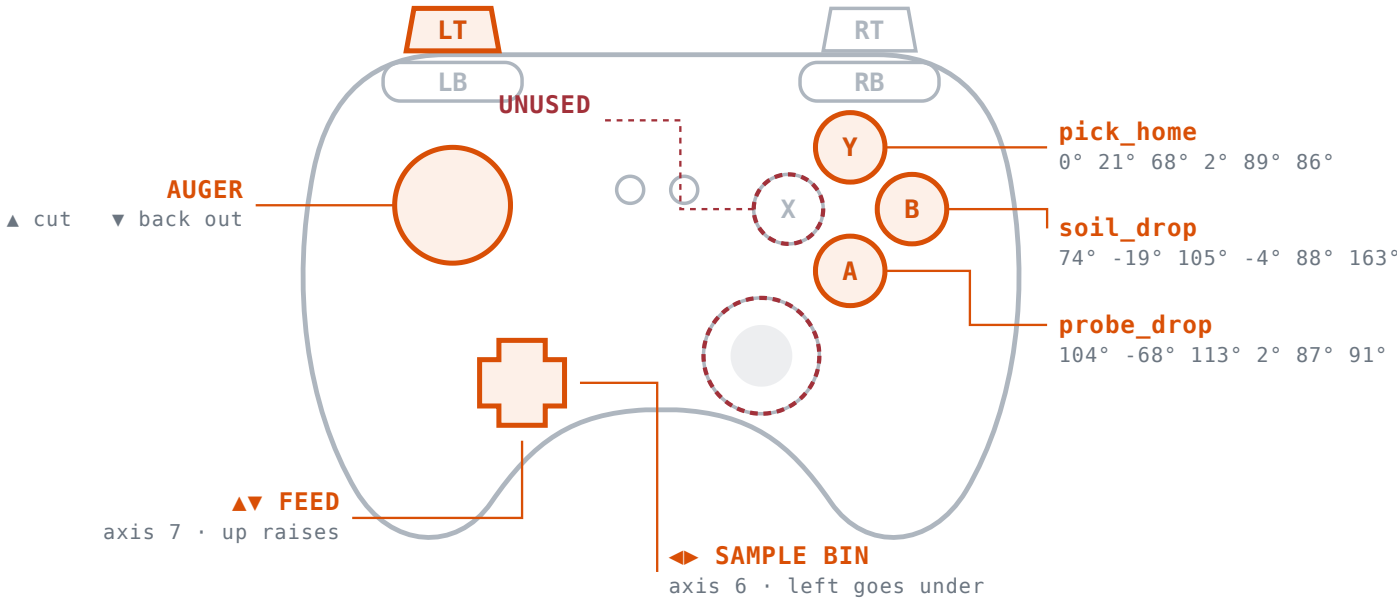
RB wins. A brushed trigger can't change the sticks.

TRIGGER THRESHOLD

LT / RT engage past **0.50**, release below **0.35**.

SIGNAL LOSS

/joy silent **0.35 s** → arm stops.



LT carries two things that never touch each other: the face buttons plan an arm move, the d-pad and left stick drive the drill.

STORED POSES – PRESS, DON'T HOLD

Y	pick_home 0° 21° 68° 2° 89° 86°
A	probe_drop 104° -68° 113° 2° 87° 91°
B	soil_drop 74° -19° 105° -4° 88° 163°
X	Unassigned.
Refused if	RB, RT or LB held · mid-move · two at once. logged, never silent

DRILL FEED & BIN – D-PAD

▲ ▼	Feed carriage axis 7 · up raises
Travel	–0.375 to +0.185 m 0.05 m/s
◀ ▶	Sample bin axis 6 · left goes under
Travel	–0.1304 to 0 m –0.1304 = on the bit axis
Stops	Head on the base plate, carriage on the mast plate.

AUGER – LEFT STICK

Stick ▲	Clockwise from above · cutting
Stick ▼	Anticlockwise backing out
Rate	Up to ±30 rad/s ~285 rpm · deadzone 0.20
Centre	Spindle stops. No end stops either way.
Release LT	Auger stops, feed stays put.

LB blocks the drill. LB + LT reaches nothing — the rover's drive enable wins over the drill's gate exactly as it does over the arm, so the rover can never drive off with the auger spinning and the mast down. Blocking behaves like releasing LT: auger to zero, feed latched. **The up-stroke is what lets the bin pass under** — clearing the rim needs 0.1787 m of the mast's 0.185, so the order is feed down, retract, bin left, dump. Feed targets follow /joint_states while LT is off, so the next press resumes from where the drill actually is.

CANONICAL LAYOUT

D-pad is **axis 7** +1 up, **axis 6** +1 left.

TRIGGERS ARE AXES

LT = axis 2, RT = axis 5, **0.0** released to **1.0** pressed.

MAPPING & LIMITS

gamepad.yaml, joystick.yaml, drill.xacro

DRILL IS SIM ONLY

No drill driver on the real rover yet. Poses: aries.srdf